



Assessment Brief

Module Code	Module Title
CSE5014	Business Analytics
Academic Year	Semester
2025	4
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Assessment Details

Assessment title	Abr.	Weighting
Application of Business Analytics in Decision Making in Telecommunication Service Sector in Sri Lanka.	WRIT1	100 %
Pass marks are 40% for undergraduate work and 50% for postgraduate work unless stated otherwise.		

Task/assessment brief:

Version 1: Introduction

The Business Analytics subject domain is considered to be one of the major area where most of the companies and various profitable and non- profitable institutions consider for achieving the best decision support in their respective various operations life cycles. Because of the economy is growing rapidly to achieve tangible and intangible as well as financial and non-financial benefits all most all the government and private organizations required to consider precision and accuracy of their management decisions in all major three managerial levels; operational, tactical and strategic. The amount of information generated in modern agile economic environment is very high and maintaining the consistency also a challenge. Because of the fact that utilization of big data analysis considered as one of the prime concern to deal with credible and valuable informed decision making. During the process of conversion data in to decision supportive information, it is very important to use different data analysis methods, techniques and tools that are comprehensively explained in data analytics. The inclusion of data analytics subject elements in modern management information systems and decision support information systems for enabling effective and efficient online analytical processing incorporated with connected operational databases, data marts and data warehouses considered to be vital.

In performing big data analysis, it is very important to use good statistical software. At present there are many such products available under generic or bespoke software category considering open source or closed source. Usually open source products are financially feasible for many organizations compared to closed source products. At present big data analysis rapid growth identifiable in open source category with frequently released updated versions and many feature extensions compared to closed source. On the other hand, side much reliable many software products have been released by industry pioneer solution providers. Therefore, section of the best product for data analysis is also need to be done wisely by relevant authorities of organizations for their objectives to be precisely achieved.

Scenario

The telecommunication industry is considered one of the most profitable and significant contributors to global economic and technological development. The strength of telecommunication services is regarded as a key indicator of a country's level of economic advancement. These services are essential for industrial development and for improving people's quality of life by enabling seamless global communication.

Due to the increasing demand for telecommunication services, numerous companies have emerged worldwide to offer both local and international services. In providing these services, improving customer satisfaction and reducing customer attrition (or churn) are vital for maintaining a loyal customer base and achieving sustainable business growth. This research aims to examine several major factors that influence customer attrition rates. These factors include monthly charges, data usage, call minutes, customer tenure (in months), Arial MT, and customer type.

As one of the developing countries in Asia, Sri Lanka places high importance on efficient and high-quality telecommunication services, which directly support the nation's GDP and per capita income. Accordingly, the government promotes a competitive market environment for telecommunication service providers (Telcos), offering greater benefits to customers.

Assume you are a senior data analyst at one of the leading Telcos in Sri Lanka. You have been assigned to develop statistically significant decision models related to customer attrition. You are required to complete this task using the provided dataset and to engage in critical discussions based on your findings. The Data Dictionary of **Telecom_Info.csv** can be seen below

Variable Name	Data Type	Description
MonthlyCharges	Float	Average monthly bill charged to the customer in USD. Range: ~20–120.
DataUsageGB	Float	Monthly mobile data usage by the customer in gigabytes. Correlated with charges.
CallMinutes	Float	Total number of call minutes used by the customer per month.
CustomerTenureMonths	Float	Number of months the customer has been subscribed to the service.
NumComplaints	Float	Number of complaints the customer has filed recently. Higher values may indicate dissatisfaction.
ChurnProbability	Float	Target variable. Predicted probability (0 to 1 scale) of the customer churning.
Customer Type	Categorical	Type of customer: - Individual - Business - VIP

Tasks

1. Describe with credible examples of possible advantages generated by analytics and business intelligence found in data science for various levels of officials in the aforesaid organization to facilitate the **Vision** for effective, efficient and suitable informed decisions making in Air Transportation sector of the country. **(5Marks)** LO2
 2. Explain tools, techniques and methodologies going to use for **this** case study based analysis **(5Marks)** LO1
 3. Use central tendency analysis for *Monthly Charges, Data Usage GB, Call Minutes, Customer Tenure Months* and **Churn Probability**. Represent finding graphically using bell curves. The answers should be followed by descriptive justifications. **(15 Marks)** L O 3
 4. Use statistical hypothesis analysis to determining whether vehicle **Churn Probability** significantly fluctuated **type of the customer**. Statistical analysis should be comprised of numerical findings followed by graphical analysis. Discuss the advantages generated through the analysis for informed decision making. **(15 Marks)** LO3
 5. Using statistical hypothesis testing, prove whether there are statistically significant relationships exists on **ChurnProbability** with other features such as **MonthlyCharges, Data UsageGB, CallMinutes, CustomerTenureMonths, NumComplaints** and **ChurnProbability** of the aircraft. The answers should be followed by descriptive justifications based on inferential statistical findings. **(35 Marks)** LO3
 6. Write a conclusion followed by recommendation based on the findings of the data analysis and suggest necessary recommendations as the solution for the problems identified. The answers should be followed by descriptive justifications. **(15 Marks)** LO2 LO3
 7. Proper report format and standard scientific report writing. **(10 Marks)** LO1
- Question no 5 should be incorporated with hypothesis based **normality testing**.
 - The conclusion can contain the findings of suitable the **regression analysis** apart from other findings.
 - Students are expected to do the assignment tasks with the support of provided **Aviation_Info.csv** dataset.
 - Students are strictly required to submit their **R project file, R script file** and suitable evidences of own work along with their report. Failure to do so would disqualify their report for marking.

Word count (or equivalent):

3000

This a reflection of the effort required for the assessment. Word counts will normally include any text, tables, calculations, figures, subtitles and citations. Reference lists and contents of appendices are excluded from the word count. Contents of appendices are not usually considered when determining your final assessment grade.

Academic or technical terms explained:

- **BA**- Business Analytics done through statistical tools, techniques and methods
- **BI**-Business Intelligence generated through BA this encompasses narrative descriptions, Graphs and Formula's
- **Describe**- Demonstrate an understanding of the facts based on previously learned information in BA module
- **Hypothesis**: Null and alternative hypothesis written for supporting research questions for statistical
- **Use** - Apply BA knowledge to actual situations to generate new knowledge
- **Explain**- Demonstrate an understanding of the facts based on the BA and the scenario.
- **Analyze** – Break down data into simpler parts and find evidence to support generalizations.
- **Discuss** - Demonstrate an understanding of the facts f the facts based on the BA and the scenario along with suitable real world examples.
- **Justify** - Make and defend judgments based on internal evidence through findings of the statistical analysis
- **Conclude**- Make and defend judgments based on internal evidence through findings of the statistical analysis

Submission Details

Submission Deadline:	This will be provided on the Moodle submission point.	Estimated Feedback Return Date	This will normally be 20 working days after initial submission.
Submission Time:	By 2.00pm on the deadline day.		
Moodle/Turnitin:	Any assessments submitted after the deadline will not be marked and will be recorded as a non-attempt unless you have had an extension request agreed or have approved mitigating circumstances. See the School Moodle pages for more information on extensions and mitigating circumstances.		
File Format:	The assessment must be submitted as a pdf document (save the document as a pdf in your software) and submit through the Turnitin submission point in Moodle. Your assessment should be titled with your: student ID number, module code and assessment ID, e.g. st12345678 BHL5007 WRIT1		
Feedback	Feedback for the assessment will be provided electronically via Moodle. Feedback will be provided with comments on your strengths and the areas which you can improve. View the guidance on how to access your feedback. All marks are provisional and are subject to quality assurance processes and confirmation at the programme Examination Board.		

Assessment Criteria

Learning outcomes assessed

- L1: Explain Business Analytics methodologies, tools and the techniques.
- L2: Evaluate business advantages produced by business analytics
- L3: Develop propose solution for a business problem or creating opportunity using appropriate business analytics methodologies, tools and the techniques.

Other skills/attributes developed

This includes elements of the Cardiff Met EDGE (Ethical, Digital, Global and Entrepreneurial skills) and other attributes developed in students through the completion of the module and assessment. These will also be highlighted in the module guidance, which should be read by all students completing the module. Assessments are not just a way of auditing student knowledge. They are a process which provides additional learning and development through the preparation for and completion of the assessment.

ETHICAL	Understanding the importance of adhering to the formal ethical practices when different sources data is extracted, processed and disseminate. This is achieved via university ethical guidelines and procedures.
DIGITAL	Make use of data collected or provided for cleansing, formatting, analyzing, processing and signifying statistically with the support of statistical tools, techniques and methods.
GLOBAL	Usage of extracted data related to various parts or scenarios of the world for statistical analysis and processing. Localizing findings as per requirements of organizations and individuals.
ENTREPRENEURIAL	Involvement of business analytics and business intelligence; motivate students to do self-reflective work as informed decision makers of their own organizations belongs to different industries.

Task	Poor < 40	Satisfactory 40 - 54	Good 55 -69	Excellent 70 -100
1	Did not Explain the scenario and not included the benefits provided by the analysis for the selected company/institution.	Explained the scenario but no benefits provided by the analysis for the selected company/institution.	Explained the scenario and benefits provided by the analysis for the selected company/institution.	Well explained the scenario and the benefits provided by the analysis for the selected company/institution. Examples included. Citations included.
2	Did not state tools, techniques and methodologies going to be used for analysis	Stated tools, techniques and methodologies going to be used for analysis	Stated tools, techniques and methodologies going to be used for analysis. Explained the mentioned tools, techniques and methodologies	Stated tools, techniques and methodologies going to be used for analysis. Well explained the mentioned tools, techniques and methodologies.
3	Did not include bell curves based on data provided	Included bell curves based on data provided	Included bell curves based on data provided and Described information represented in the charts briefly	Included bell curves based on data provided and Well described information represented in the charts briefly

4	Did not include hypothesis statement considering null hypothesis and alternative hypothesis	Included hypothesis statement considering null hypothesis and alternative hypothesis and statistical justification of result based on most suitable mean or variance analysis tests.	Included hypothesis statement considering null hypothesis and alternative hypothesis and statistical justification of result based on most suitable mean or variance analysis tests supported by basic discussion and graphical analysis supported by basic discussion.	Included hypothesis statement considering null hypothesis and alternative hypothesis and statistical justification of result based on most suitable mean or variance analysis tests supported by excellent discussion and graphical analysis.
5	Did not include of hypothesis statement considering null hypothesis and alternative hypothesis	Included hypothesis statement considering null hypothesis and alternative hypothesis and statistical justification of result based on correlation analysis test	Included hypothesis statement considering null hypothesis and alternative hypothesis and statistical justification of result based on correlation analysis test supported by normality test. Graphical analysis also included.	Included hypothesis statement considering null hypothesis and alternative hypothesis and statistical justification of result based on correlation analysis test supported by normality test and the regression test. Graphical analysis also included.
6	Did not provide conclusion related to statistical findings done based on hypothetical testing.	Provided conclusion related to statistical findings done based on hypothetical testing.	Provided conclusion related to statistical findings done based on hypothetical testing. The conclusion should comprise of sub conclusions of question no 04 and 05.	Provided conclusion related to the findings done based on statistical tests The conclusion should comprise of sub conclusions of question no 04 and 05. The student provided recommendation to support improvements of the organization. Applications of suitable regression analysis findings are highly anticipated.
7	No or weak report structure adhered.	Standard report structure included but no scientific reporting style used to explain the findings.	Standard report structure included and scientific reporting style used to explain the findings. References included but have some errors.	Standard report structure included and scientific reporting style used to explain the findings. Proper citation with matching references included.

Who can answer questions about my assessment?

Questions about the assessment should be directed to the staff member who has set the task/assessment brief. This will usually be the Module Leader. They will be happy to answer any queries you have.

Staff members can often provide feedback on an assignment plan but cannot review any drafts of your work prior to submission. The only exception to this rule is for Dissertation Supervisors to provide feedback on a draft of your dissertation.

Referencing and independent learning

Please ensure you reference a range of credible sources, with due attention to the academic literature in the area. The time spent on research and reading from good quality sources will be reflected in the quality of your submitted work.

Remember that what you get out of university depends on what you put in. Your teaching sessions typically represent between 10% and 30% of the time you are expected to study for your degree. A 20-credit module represents 200 hours of study time. The rest of your time should be taken up by self-directed study.

Unless stated otherwise you must use the **HARVARD** referencing system. Further guidance on referencing can be found in the Study Smart area on Moodle and at www.citethemrightonline.com (use your university login details to access the site). Correct referencing is an easy way to improve your marks and essential in achieving higher grades on most assessments.

Technical submission problems

It is strongly advised that you submit your work at least 24 hours before the deadline to allow time to resolve any last minute problems you might have. If you are having issues with IT or Turnitin you should contact the IT Helpdesk on (+44) 2920 417000. You may require evidence of the Helpdesk call if you are trying to demonstrate that a fault with Moodle or Turnitin was the cause of a late submission.

Extensions and mitigating circumstances

Short extensions on assessment deadlines can be requested in specific circumstances. If you are encountering particular hardship which has been affecting your studies, then you may be able to apply for mitigating circumstances. This can give the teachers on your programme more scope to adapt the assessment requirements to support your needs. Extensions and mitigating circumstances policies and procedures are regularly updated. You should refer to your degree programme or school Moodle pages for information on extensions and mitigating circumstances.

Unfair academic practice

Cardiff Met takes issues of unfair practice **extremely seriously**. The University has procedures and penalties for dealing with unfair academic practice. These are explained in full in the University's Unfair Practice regulations and procedures under Volume 1, Section 8 of the Academic Handbook. The Module Leader reserves the right to interview students regarding any aspect of their work submitted for assessment.

Types of Unfair Practice, include:

Plagiarism, which can be defined as using without acknowledgement another person's words or ideas and submitting them for

assessment as though it were one's own work, for instance by copying, translating from one language to another or unacknowledged paraphrasing. Further examples include:

- Use of any quotation(s) from the published or unpublished work of other persons, whether published in textbooks, articles, the Web, or in any other format, where quotations have not been clearly identified as such by being placed in quotation marks and acknowledged.
- Use of another person's words or ideas that have been slightly changed or paraphrased to make it look different from the original.
- Summarising another person's ideas, judgments, diagrams, figures, or computer programmes without reference to that person in the text and the source in a bibliography/reference list.
- Use of assessment writing services, essay banks and/or any other similar agencies (NB. Students are commonly being blackmailed after using essay mills).
- Use of unacknowledged material downloaded from the Internet.
- Re-use of one's own material except as authorised by your degree programme.

Collusion, which can be defined as when work that has been undertaken with others is submitted and passed off as solely

the work of one person. Modules will clearly identify where joint preparation and joint submission are permitted, in all other cases they are not.

Fabrication of data, making false claims to have carried out experiments, observations, interviews or other forms of data collection and analysis, or acting dishonestly in any other way.

How is my work graded?

Assessment grading is subject to thorough quality control processes. You can view a summary of these processes on the [Assessment Explained Infographic](#).

Grading of work at each level of Cardiff Met degree courses is benchmarked against a set of general requirements set out in [Volume 1, Section 4.3](#) of our Academic Handbook. A simplified version of these Grade Band Descriptors (GBDs) with short videos explaining some of the academic terminology used can be accessed via the [Facilitation of Learning](#) resource page.

We would strongly recommend looking at the [Study Smart](#) area of Moodle to find out more about assessments and key academic skills which can have a significant impact on your grades. Always check your work thoroughly before submission.

Cardiff Met

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